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RESEARCH ARTICLE

BGL-PhishNet: Phishing Website Detection Using Hybrid Model-BERT, GNN, and LightGBM

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ABSTRACT Phishing attacks exploit human and technological vulnerabilities to steal sensitive information, posing a significant threat to online security. To address this challenge, this research introduces a new hybrid method for phishing detection using three advanced techniques: BERT for analyzing texts, Graph Neural Networks for analyzing URLs, and LightGBM for extracting key metadata features. The hybrid approach combines these techniques into a multi-layered model to improve detection accuracy by focusing on text, URL structure, and metadata. The hybrid model achieves an accuracy of 97.3%, outperforming current state-of-the-art models. Experimental results show that this method reduces false positives and enhances phishing detection across various online platforms. The system ensures better real-time performance by integrating outputs from BERT, GNNs, and LightGBM. It also addresses evolving phishing tactics by considering both lexical and structural URL features. This multi-level approach highlights the importance of strengthening online security through robust detection mechanisms.

INDEX TERMS Phishing, URL detection, residual pipelining, cybersecurity, classification, agriculture.

I. INTRODUCTION

A. PHISHING ATTACK DETECTION

In 2022, phishing attacks accounted for nearly 36% of global data breaches, causing an estimated financial loss of \$12 billion annually. These attacks exploit human vulnerabilities and technological gaps, often masquerading as legitimate entities to deceive users into divulging sensitive information. As digital platforms increasingly dominate communication and commerce, phishing threats grow in complexity and frequency, underscoring the urgent need for robust and adaptable detection systems [1], [2].

Phishing detection has evolved from rule-based and heuristic methods to machine learning (ML) and deep learning

(DL) models, which offer improved accuracy and adaptability. Despite these advances, traditional detection systems face significant limitations when dealing with sophisticated and dynamic phishing tactics.

Phishing websites copy recognizable web pages by modifying URL patterns, adding text, and modifying metadata. Because existing detection systems tend to be one-dimensional, they are often ineffective in making a distinction between these. Rule-based techniques are not versatile for novel types of phishing, as with ML approaches that use CNNs and SVMs [3]. Even then, there are challenges such as real-time functioning, scaling, and generalization related to disparate data settings. More so, high false-positive rates diminish the credibility and suitability of such mechanisms.

Historically, phishing detection systems implemented either heuristic or signature detection methods. While some

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of these models can effectively defend against known threats, they fail to keep pace with the constantly changing methods of attackers. Many bogus deception pages are constructed to resemble existing pages, escaping detection by ordinary systems. Machine learning-based methods have been introduced to identify patterns and detect abnormal characteristics of phishing websites, but even these approaches struggle with URLs that closely mimic legitimate ones. Addressing these challenges requires advanced detection strategies that employ multiple types of information to extract diverse features [4], [5].

The hybrid model adopts a novel multimodel approach. BERT analyzes language structure to detect elements signaling phishing, GNNs examine URL structures as graphs to uncover deviations and LightGBM evaluates metadata such as domain registration and SSL certificates. By integrating these perspectives, the model enhances detection accuracy and reduces false positives. Experimental results demonstrate that the hybrid model achieves 97.3% accuracy, 97.8% precision, 96.7% recall, and an AUC score of 0.97, significantly outperforming state-of-the-art methods. This innovative system highlights the importance of multi-faceted detection strategies to address the evolving phishing landscape.

B. MOTIVATION

In addition, the increasing frequency of phishing attempts also has serious implications for personal privacy, financial security, and organizational reputation. With the ubiquitousness of the Internet and on-line transactions, phishing schemes have matured, and threats posed through phishing attacks have extended to exploiting user behavior and technology. Conventional detection strategies are usually ineffective against these developments due to their statics, rule-based approaches, and rigidity to new strategies.

This indicates that alternative approaches are vital to addressing the problem of phishing. ML techniques have been noticed to solve such problems. The benefits of various models including BERT, GNNs, and LightGBM are concerted in the hybrid system to improve its accuracy and scalability. Phishing threats can be effectively tackled through the collaborative learning capabilities of the model, improving the safety of people and organizations on the Internet.

The remaining sections are organized as follows: Section II presents a detailed review of related works, discussing the evolution of phishing detection methods, and Section III identifies key research gaps. The challenges of the existing models and key contributions of the hybrid approach are described in Sections IV and V respectively. Section VI represents the materials and methods. Sections VII, VIII, and IX describe the hybrid methodology, including data preprocessing, feature extraction, model architecture, and mathematical representations. Section X outlines the experimental setup and datasets, focusing on evaluation metrics and benchmarks. This section discusses the results, compares the performance of the hybrid model with existing approaches, and demonstrates its effectiveness. Finally, Section XI concludes with

a summary of findings, implications, and future research directions.

II. RELATED WORKS

Phishing detection is an area that is constantly changing and developing with the implementation of different techniques. These techniques include hybrid methods that combine various detection mechanisms, state-of-the-art deep learning architectures, conventional machine learning techniques, and feature-based algorithms. In the following sections, most of the contributions in these domains are addressed by comparing the benefits of detection accuracy, complexity, flexibility, and scalability.

A. TRADITIONAL MACHINE LEARNING METHODS

Simplicity and ease of use keep traditional ML models like Support Vector Machines (SVMs), Random Forests, and K-Nearest Neighbour (KNN) are relevant. Abusaimh et al. [6] demonstrate the efficiency of SVM in binary classification for anti-phishing systems, though its tendency to overfit and need for frequent retraining limits its scalability for evolving phishing threats. Sinha et al. [20] focus on Random Forest classifiers, highlighting their efficiency in handling imbalanced datasets through ensemble approaches to improve detection rates. However, their slower expression makes them less suitable for scenarios requiring immediate responses. Kalabarige et al. [7] underscore KNN's simplicity and speed, which work well in small-scale systems but face challenges with memory and processing as data scales, limiting its effectiveness in large datasets. Ali and Zaharon [8] explore ensemble learning methods such as bagging and boosting, which enhance accuracy but increase computational complexity, which pose challenges for real-time phishing detection due to the demand for higher processing power and time.

B. DEEP LEARNING-BASED TECHNIQUES

Deep learning methods like recurrent neural networks (RNNs) and convolutional neural networks (CNNs) have shown significant success in phishing detection due to their ability to learn complex data patterns automatically. Ripa et al. [9] highlight RNNs' enhanced training speed with GPU computation, though the black-box nature of RNNs poses transparency issues in cybersecurity. CNNs, as emphasized by Sánchez-Paniagua et al. and Huang et al. [10], [11], are effective in URL and email phishing detection by extracting complex features but face challenges due to high computational requirements and data demands. Yang et al. [12] applied deep learning to detect phishing in social media, though the time-consuming training limits real-time applicability. Zhou et al. [13] used deep reinforcement learning to adapt to new phishing methods, but the high resource costs make it impractical for many real-world applications.

C. FEATURE-BASED AND RULE-BASED METHODS

Feature extraction and rule-based systems are common strategies in phishing detection. Mohammad et al. [30] emphasize

feature extraction from URLs and webpages to enhance detection, highlighting that feature selection significantly impacts model accuracy. However, this approach struggles in dynamic environments due to its reliance on manual feature selection and domain knowledge. Tan et al. and Tang and Mahmoud [14], [15] focus on rule-based systems that apply predefined rules for identifying phishing attempts. While these systems are simple and useful for small-scale applications, they lack adaptability to evolving phishing tactics, necessitating advancements to enhance flexibility and threat responsiveness.

D. HYBRID APPROACHES

Hybrid models combine various detection approaches to improve phishing detection. Geest et al. [16] propose a model integrating ML techniques, rule-based methods, and heuristics to enhance precision by leveraging ML's adaptability, the efficiency of rules, and basic pattern recognition. However, the complexity of these systems can hinder integration, scalability, and real-time usage. Abutair and Belghith [17] highlight the need for extensive tuning in hybrid models combining ML and self-generated rules, which complicates implementation amid evolving phishing tactics. Sahingoz et al. [18] introduce a layered hybrid system that maximizes detection through redundancy but faces challenges in real-time deployment due to high processing demands.

E. MULTI-MODAL AND CLUSTERING APPROACHES

Recent strategies integrate multi-modal data and clustering methods for phishing detection. Faris and Yazid [19] propose a multi-modal DL model leveraging both visual and text data, enhancing detection by examining webpage content and layout. However, its high resource demands limit applicability in low-resource settings. He et al. [13] employ DL for semantic analysis, improving detection but facing challenges with model complexity and extended training times, hindering real-time use. Sinha et al. [20] utilize clustering techniques for phishing email detection, which works well for small datasets but loses accuracy with larger datasets, necessitating frequent retraining. Miyamoto et al. [21] focus on domain analysis using ML, which struggles with evolving phishing domains.

This diverse range of phishing detection techniques highlights the complexity of the problem, as each approach comes with its strengths and limitations. Hybrid models offer robustness but are computationally intensive, while DL techniques push the boundaries of detection accuracy at the cost of scalability and resource demands. Traditional ML methods provide efficient solutions for static environments, though they may falter in real-time and dynamic scenarios. Feature-based and rule-based systems offer simplicity but require constant updates to keep pace with evolving threats. Finally, multi-modal and clustering approaches demonstrate promise, though their computational requirements and precision in large-scale datasets remain challenges. The summary

of the state-of-the-art methods is presented in Table 1, which includes key strengths, major limitations, real-time detection suitability, and scalability for each method.

III. RESEARCH GAP

Phishing website detection faces challenges in adapting to sophisticated attacks. Although traditional methods like ML, rule-based, and heuristic approaches have been effective, they struggle with advanced phishing schemes. Integrating models like deep learning, GNN, and ensemble methods holds potential but requires more research to understand their combined efficacy.

While models like Random Forests and SVMs perform well on complex problems, they face issues like slow prediction and frequent retraining. GNNs and BERT offer solutions, but their real-time application needs optimization. Most studies lack focus on practical, user-friendly phishing detection tools that integrate seamlessly into daily activities, emphasizing the need for intuitive solutions and user education.

In summary, the primary research gap in phishing website detection lies in the need for:

- 1) More comprehensive studies are needed that focus on the synergistic integration of advanced detection methodologies, including machine learning, rule-based systems, deep learning models like BERT, structural analysis through GNNs, and ensemble techniques such as LightGBM.
- 2) Addressing the scalability and real-time processing challenges faced by existing detection systems is essential. Developing methods that enable advanced hybrid models to efficiently handle large datasets and provide rapid predictions in dynamic environments remains a significant challenge.
- 3) There is a critical need for the development of user-centric tools and solutions that deliver effective phishing detection for end-users. These tools should empower users with real-time protection and easy integration into their online activities while adapting to evolving phishing tactics.
- 4) Further exploration is warranted on the integration of behavioral analysis into phishing detection frameworks. Understanding user interaction with phishing attempts can provide context and enhance the robustness of detection mechanisms.

By addressing these research gaps, the hybrid phishing detection model promises to pave the way for more resilient, scalable, and user-centric phishing detection solutions that are better equipped to tackle the rapidly evolving landscape of online threats.

IV. CHALLENGES IN PHISHING WEBSITE DETECTION

Phishing website detection presents numerous challenges that hinder the effectiveness of existing security systems in combating increasingly sophisticated attacks. One of the primary difficulties is that attackers continuously adapt their tactics to evade detection, employing advanced tools and

TABLE 1. Comparison of phishing detection methods.

Method	Key Strengths	Major Limitations	Real-Time Detection Suitability	Scalability
Hybrid Approaches (R.J. van Geest et al., 2017; Abu-Nimeh et al., 2017; Buber et al., 2019) [6] [7], [8]	Combines the strengths of multiple methods (e.g., machine learning, rule-based, heuristic)	High complexity, integration challenges, and scalability issues	Moderate - Needs optimization for real-time	Low - Resource-intensive, hard to scale
Recurrent Neural Networks (RNNs) (Lizhen Tang et al., 2020) [9]	Fast training with GPU optimization, suitable for dynamic data	Lack of interpretability (black-box nature)	High - Suitable for real-time detection	Medium - Requires high computational resources
Convolutional Neural Networks (CNNs) (Wesam Al-Nabki et al., 2020; Huang et al., 2021) [10] [11]	Excellent at feature extraction from URLs and emails, highly accurate	High computational costs, needs large datasets	Medium - Computational demands limit real-time use	Low - Resource-intensive, hard to deploy at scale
Support Vector Machines (SVMs) (Wang et al., 2019) [12]	High accuracy in binary classification tasks, good for small datasets	Struggles with large datasets, requires frequent retraining	Medium - May need retraining for evolving threats	Low - Not ideal for large, dynamic datasets
Random Forests (Zhou et al., 2019) [13]	Handles imbalanced datasets well, good generalization	Slower prediction speeds, less suitable for real-time use	Low - Slow to generate results in real-time	Medium - Handles larger datasets but at a slower speed
K-Nearest Neighbor (KNN) (Abdelhamid et al., 2014) [14]	Simple, quick to implement, effective in small environments	Struggles with scalability and large datasets	Medium - Can perform well in small environments	Low - Inefficient in handling large datasets
Ensemble Learning (Chen et al., 2018) [15]	Improved accuracy by combining models, robust	Increases computational overhead and complexity	Low - Computationally expensive for real-time	Medium - Can scale with optimization but expensive
Feature-Based Extraction (Mohammad et al., 2014) [30]	Manual feature selection enhances phishing detection accuracy	Requires domain expertise, not adaptable to new phishing tactics	Medium - Requires constant updates, limiting real-time use	Low - Needs manual intervention, hard to scale
Rule-Based Systems (Feroz and Mengel, 2015; Chiew et al., 2015) [17] [18]	Simple to implement, straightforward	Static nature, ineffective against evolving phishing tactics	High - Effective in small, controlled environments	Low - Inflexible, poor adaptability to new threats
Multi-Modal Deep Learning (Faris et al., 2020) [19]	High accuracy by integrating visual and textual data	Requires large datasets and high computational power	Low - Computationally demanding for real-time	Low - Limited scalability due to resource needs
Clustering Techniques (Sinha et al., 2020) [20]	Lightweight and fast for small datasets, effective for quick detection	Lacks precision with large datasets, needs frequent retraining	Medium - Effective in small datasets but weak with large data	Low - Struggles with scaling to large environments
Deep Reinforcement Learning (Zhou et al., 2018) [21]	Adapts dynamically to new phishing tactics	Very resource-intensive, complex to implement	Low - Not feasible for real-time without significant resources	Low - Requires heavy infrastructure for large-scale use
Naive Bayes (Basnet et al., 2012) [22]	Computationally efficient, fast classification	Assumes feature independence, reducing accuracy in complex data	High - Well-suited for real-time use in simple cases	Medium - Efficient but less accurate for larger datasets

techniques that allow them to circumvent traditional antivirus and firewall protections. This ongoing evolution in phishing strategies necessitates a responsive and adaptive detection framework that can dynamically adjust to new threats.

Another well-established technique which was noted being used by attackers is the making of misleading subdomains. This issue makes it hard for users to know the difference between genuine and dangerously malicious sites. With respect to phishing more generally, this concept aggravates a recourse by users towards the phishing websites, which is quite high and underlines the necessity of more advanced detection systems than currently available to handle websites scope and tones accurately. In this context, it is precisely on us that GNNs can be especially useful since they will be able to consider the relationships between URLs and look for small traces of phishing.

A category of attacks that comes with time constraints, for instance, the Watering Hole Attack. This method causes increased difficulty in coming up with detection systems. As specific types of assaults tend to take place over a shorter

period, they pose a problem for security experts trying to hunt them down and eliminate them quickly. These issues are again addressed by developing mechanisms for real time detection systems which are able to counter new threats as they arise. Such models can be effectively applied in detecting interactive attacks by utilizing BERT in text understanding and graph neural networks in understanding the structure [22].

In addition, there is lack of adequate information which increases the problems of identifying phishing sites. Factors determining the efficiency of any detection algorithms include availability of datasets for algorithm training and testing. The issue of insufficient data slows down the development and assessment of effective detection systems, thus affecting the progress in the field. This challenge emphasizes the need to address this deficiency by combining different approaches including techniques such as LightGBM that can process structured data and metadata in the detection system.

False positives also present more problems as genuine pages may be upto impersonation inducing attack warnings. This can result in the anti-phishing technology being

ineffective due to people's loss of confidence towards the systems, and hence reduces effectiveness of user experience. As people get exposed to fake alerts that go more than three instances, their trust in detection systems gets eroded which ultimately affects their behavior online exposing them to real phishing executions. The hybrid system that is being suggested will focus on reducing false positives using various feature extraction approaches and a refined understanding of phishing indicators.

Also, the real-time anti-phishing techniques explained the importance of fast-transforming computing architectures and efficient algorithms excellent at large volumes of data processing. The application of semi-supervised and online learning in the proposed systems compensates for the associated high costs by ensuring that new threats are less computationally intensive to detect. It is in such cases where emerging technology will arise where efficient algorithms that will combat phishing attacks and yet not put a drain on the system will be developed and deployed.

Thus, given these research gaps, it is also important to formulate approaches to the above-stated problems in order to contribute to research on phishing detection. This improving data availability, and creating effective detection systems in real time, thereby allowing the creation of stronger systems that can withstand developments in phishing attacks more efficiently. These innovations will greatly improve the efficiency of mechanisms that detect phishing, mitigating the negative effects of cyber crimes on people and organizations in the modern world.

V. KEY CONTRIBUTIONS OF THE SUGGESTED APPROACH

The hybrid approach to phishing website detection presents several key contributions addressing existing challenges, research gaps, and insights from the literature review. These contributions enhance the overall effectiveness of phishing detection systems and advance the field as follows:

- 1) This approach integrates BERT, GNNs, and LightGBM into a cohesive framework, leveraging their combined strengths for enhanced detection accuracy and resilience against evolving phishing strategies.
- 2) The phishing detection framework efficiently addresses real-time resource demands, enhances scalability, and swiftly processes large data volumes to detect emerging threats.
- 3) Improves detection accuracy and reduces false alarms by leveraging advanced feature extraction and extensive training data, enhancing user trust in anti-phishing software.
- 4) Focuses on user-friendly tools like browser extensions and mobile apps to empower users and organizations in proactively identifying and avoiding phishing threats.
- 5) Aims to protect information system users by implementing dynamic, continuously learning mechanisms to prevent phishing attacks while ensuring seamless integration with existing measures.

- 6) Addresses data limitations by employing data augmentation techniques to enhance availability and diversity, improving model performance.

VI. MATERIALS & METHODS

A. MATERIALS

1) DATASET

Within this research, we have chosen the dataset labelled "Phishing Website URLs" sourced from Kaggle and consists of 549,346 URLs. Such a dataset is particularly valuable for training machine learning algorithms, as it includes a wide variety of both legitimate and phishing URLs, making it comprehensive and representative of real-world scenarios. In the dataset, every URL is marked either for legitimate pages (0) or phishing pages (1), which serves as a target variable for modeling.

To better understand the characteristics and feature distributions of dataset, t-SNE visualization was applied to the extracted features. This dimensionality reduction technique revealed the underlying structure of the data, showcasing distinct clusters for phishing and legitimate URLs. This analysis not only validated the quality of the dataset but also informed further feature engineering, ensuring that the dataset was well-prepared for model training and enhancing the robustness of the phishing detection system.

- URL: This column contains the actual web addresses, which are critical for analysis. The structural elements of these URLs, including subdomains, query parameters, and paths play a significant role in identifying phishing sites. Phishing URLs often exhibit deceptive structures that mimic legitimate ones, thus complicating detection efforts.
- Label: This binary target variable indicates whether the URL is phishing or legitimate. The labels are crucial for supervised learning approaches, enabling the model to learn distinguishing features between the two classes.

To ensure a balanced training set, we employed techniques such as oversampling of phishing URLs and undersampling of legitimate ones. This approach avoids class imbalances that could skew the model toward predicting the majority class more frequently, thereby enhancing the ability of the model to detect phishing attempts accurately.

In addition to the URL data, we extracted additional metadata features, including **domain age**, **HTTPS status**, and **WHOIS information**. These features enrich the dataset, providing the models with more meaningful insights into the characteristics of the websites being analyzed.

a: FEATURE SET

The dataset comprises a diverse mix of features categorized into lexical, host-based, and metadata features, which capture the inherent differences between phishing and legitimate websites:

b: LEXICAL FEATURES

- **URL Length:** Represents the total number of characters in the URL. Research indicates that phishing URLs tend to be longer, often incorporating additional subdomains and parameters.
- **Number of Digits and Special Characters:** Analyzing the count of digits and special characters can help identify patterns commonly associated with phishing attempts. Phishing URLs frequently use unconventional characters to obfuscate their true intent.
- **Presence of Suspicious Words:** The occurrence of words like “login,” “verify,” “secure,” and others commonly associated with phishing attacks is analyzed. These lexical cues can be critical indicators of potential phishing attempts.

c: HOST-BASED FEATURES

- **Domain Age:** The age of the domain can be indicative of its legitimacy. Phishing websites often use newly registered domains, whereas legitimate websites typically have a longer registration history.
- **Domain Registration Country:** The geographical location of the domain’s registration can provide insights into its legitimacy. Certain regions may be associated with higher rates of phishing activity.
- **Server Details:** Information about the hosting server, such as its location and configuration, can also inform the detection process.

d: METADATA FEATURES

- **WHOIS Registration Details:** These include critical information such as the creation date, expiry date, and update frequency of the domain. Newly created domains with imminent expiry dates are often red flags.
- **Presence of SSL Certificates:** Whether the website employs SSL encryption (indicated by the presence of HTTPS) is a significant factor. Legitimate websites typically use SSL certificates, whereas phishing sites may not.
- **DNS Records:** Analysis of DNS records can reveal patterns typical of phishing domains, such as irregularities in the setup or management of DNS entries.

The comprehensive feature set enables the hybrid detection model to leverage a multi-faceted approach, enhancing its ability to discern between phishing and legitimate websites effectively.

Table 2 provides a comprehensive overview of the key characteristics of the dataset used for phishing website detection. The dataset consists of a total of 549,346 URLs, which includes a balanced distribution of 40% phishing and 60% legitimate URLs, illustrating the challenge of distinguishing between the two classes. The average URL length is recorded at 72 characters, indicating the complexity often found in phishing URLs, which frequently incorporate multiple subdomains and parameters to disguise their malicious intent. Additionally, the average domain age is approximately

TABLE 2. Dataset statistics.

Statistic	Value
Total number of URLs	549,346
Distribution (Phishing vs. Legitimate)	40% Phishing, 60% Legitimate
Average URL Length	72 characters
Average Domain Age	1.5 years
Proportion of URLs with HTTPS status	65%

1.5 years, suggesting that many phishing sites are relatively new, a common trait among fraudulent domains. Lastly, the proportion of URLs with HTTPS status stands at 65%, highlighting that while a majority of legitimate sites use secure connections, a significant number of phishing sites may still lack proper SSL certification. This statistical summary emphasizes the diverse and challenging nature of the dataset, which is critical for developing effective phishing detection models.

By thoroughly analyzing this diverse dataset, the hybrid model aims to improve phishing detection accuracy and reduce false positives, ultimately contributing to a more secure online environment.

B. METHODS

A combination of BERT, GNNs, and LightGBM is insisted within the framework of an ensemble learning approach for the phishing detection system. This all-encompassing strategy seeks to fill the existing gaps or challenges that further research would take up in respect of phishing detection systems already in existence, namely flexibility, turn-around time, as well as user interaction.

1) DATA PREPROCESSING

Data preprocessing is a critical step that ensures the dataset is suitable for analysis. The preprocessing pipeline includes the following steps:

- **Cleaning:** The URLs are standardized by removing unnecessary characters, trailing slashes, and redundant subdomains such as www. This helps in eliminating noise that could hinder the detection process.
- **Tokenization:** Each URL is tokenized into its core components, such as:
 - 1) **Protocol:** Identifying whether the URL starts with HTTP or HTTPS.
 - 2) **Subdomains:** Extracting subdomains that may indicate malicious intent.
 - 3) **Domain Name:** The primary domain that needs to be assessed for legitimacy.
 - 4) **Path and Query Parameters:** Analyzing additional components of the URL that may contain suspicious elements.
- **Feature Extraction:** A diverse set of features is extracted from the URLs, categorized into three main groups, and is shown in Table 3:
 - 1) **Lexical Features**
 - **URL Length:** Total number of characters in the URL.
 - **Number of Digits and Special Characters:**

TABLE 3. Feature set overview.

Feature Type	Feature Examples
Lexical Features	URL Length, Special Characters, Suspicious Words
Host-based Features	Domain Age, Registration Country, Server Details
Metadata Features	WHOIS Information, SSL Presence, DNS Records

Counts that help identify patterns associated with phishing attempts. Presence of Suspicious Words: Occurrence of words like “login,” “verify,” or “secure.”

2) Host-based Features

- Domain Age: How long the domain has been registered; shorter ages are often indicative of phishing.
- Domain Registration Country: The geographical location of the domain registration.
- Server Details: Information about the hosting server, including its IP address and location.

3) Metadata Features

- WHOIS Registration Details: Key information such as creation and expiry dates, which help in assessing domain legitimacy.
- Presence of SSL Certificates: Indicates whether the website uses SSL encryption (HTTPS).
- DNS Records: Analysis of DNS setup can reveal patterns indicative of phishing domains.

In the pre-processing phase, t-SNE (t-Distributed Stochastic Neighbor Embedding) was employed to visualize the high-dimensional feature space of the dataset, aiding in the understanding of data distribution and inherent clustering of phishing and legitimate URLs. This step provided crucial insights into the separability of classes based on extracted features, enabling the identification of distinct clusters and areas of overlap. The t-SNE visualization helped in refining feature selection by highlighting which features contributed most significantly to class differentiation, thereby guiding subsequent model training and optimization processes. This preliminary analysis ensured that the dataset was well-structured and ready for effective model learning, enhancing the overall performance of the phishing detection system.

VII. HYBRID MODEL

The hybrid approach in this research integrates three complementary models BERT, GNNs, and LightGBM into an ensemble framework that analyzes URLs from multiple perspectives, significantly enhancing phishing detection capabilities. Each model contributes unique strengths, addressing distinct aspects of phishing URLs in terms of linguistic features, structural relationships, and metadata attributes and is described in the following sub sections.

A. BERT: LINGUISTIC FEATURE ANALYSIS

BERT is a transformer-based language model designed for analyzing textual data. In this system, BERT is fine-tuned on a labeled dataset of phishing and legitimate URLs to learn

linguistic patterns indicative of phishing attempts. Phishing URLs often include misleading text, such as embedded keywords (e.g., login, secure, or update), designed to trick users into trusting the site.

BERT processes tokenized URLs and captures the semantic relationships among tokens, providing context-aware embeddings that reflect the intent behind textual components of the URL. For example, it can differentiate between benign terms and phishing indicators even when they appear in similar contexts. This ability to extract deep semantic features makes BERT particularly effective at detecting subtle phishing tactics that evade traditional keyword-based detection methods.

In this research, the embeddings of BERT were specifically optimized to handle URL-specific patterns, such as analyzing subdomains, suspicious word placements, and encoding structures. This fine-tuning allows the model to perform robustly, even with incomplete or obfuscated URL text, improving its ability to identify phishing attempts accurately. The phishing detection problem can be defined as a binary classification task, aiming to classify each URL x_i into one of two classes: phishing (1) or legitimate (0). The dataset consists of n URLs, each represented by m extracted features:

$$X = \{x_1, x_2, \dots, x_n\}, \quad y = \{y_1, y_2, \dots, y_n\} \quad (1)$$

where $y_i \in \{0, 1\}$ is the true label for URL x_i . The feature vector for each URL is denoted as:

$$f(x_i) = [f_1(x_i), f_2(x_i), \dots, f_m(x_i)] \quad (2)$$

The BERT model operates on tokenized URLs and captures semantic patterns from the text. Let $T(x_i)$ represent the tokenized version of the URL x_i . BERT generates a high-dimensional embedding h_i for x_i , defined as:

$$h_i = \text{BERT}(T(x_i)) \quad (3)$$

Here, h_i is a contextualized vector of dimension d_B , where d_B is the embedding size of the BERT model. The BERT embeddings focus on lexical features such as suspicious keywords and text patterns.

B. GNN: STRUCTURAL FEATURE ANALYSIS

Phishing URLs often employ deceptive structural features, such as complex subdomains or mimicry of URL architectures of legitimate websites. To analyze these structural patterns, GNN model each URL as a graph. In this graph representation, nodes correspond to components of the URL (e.g., domain, subdomain, path), and edges represent relationships between these components.

The GNN processes the graph representation of URLs to identify structural anomalies that are typical of phishing websites. For instance, phishing URLs might feature unusually long paths, excessive subdomains, or uncommon relationships between query parameters and domains. By leveraging graph-based structural analysis, the GNN learns patterns and interdependencies that distinguish legitimate URLs from malicious ones.

In this study, the GNN architecture was tailored to handle complex URL structures, ensuring efficient learning and representation of graph features. The model captures both global URL properties such as overall length, hierarchy and local patterns which includes specific subdomain-path relationships, and are critical for detecting sophisticated phishing strategies.

Phishing URLs often have structural anomalies that can be modeled as graphs. Let $G(x_i) = (V, E)$ represent the graph representation of the URL x_i , where:

- V : the set of nodes representing components such as the domain, subdomain, and path.
- E : the set of edges representing relationships between these components.

The GNN learns a mapping $z_i = \text{GNN}(G(x_i))$, where z_i is a node embedding vector of dimension d_G representing the structural features of x_i . The GNN's message-passing mechanism aggregates information from neighboring nodes to identify structural anomalies. Mathematically, the message-passing update for node v in graph $G(x_i)$ can be expressed as:

$$h_v^{(l+1)} = \sigma \left(W^{(l)} \cdot \sum_{u \in \mathcal{N}(v)} h_u^{(l)} + b^{(l)} \right) \quad (4)$$

where:

- $h_v^{(l)}$: the embedding of node v at layer l ,
- $\mathcal{N}(v)$: the set of neighbors of v ,
- $W^{(l)}$ and $b^{(l)}$: the weight matrix and bias term for layer l ,
- σ : a non-linear activation function (e.g., ReLU).

After L layers, the final node embeddings z_i are aggregated to represent the overall URL structure.

C. LIGHTGBM: METADATA-BASED FEATURE ANALYSIS

LightGBM, a gradient-boosting decision tree framework, excels in processing structured metadata, making it an ideal choice for analyzing non-textual features of URLs. Phishing websites often exhibit metadata patterns that distinguish them from legitimate websites, such as shorter domain registration durations, lack of SSL certificates, or irregular WHOIS data [23], [24].

Metadata features such as domain age, SSL presence, and WHOIS data are processed by LightGBM. Let m_i denote the metadata vector for x_i . The LightGBM classifier uses a gradient-boosting mechanism to learn the decision boundary. The prediction score for x_i is computed as:

$$\hat{y}_i = \sum_{t=1}^T \eta \cdot g_t(m_i) \quad (5)$$

where:

- T : the number of decision trees,
- η : the learning rate,
- $g_t(m_i)$: the prediction from the t -th tree.

In the ensemble system, LightGBM processes metadata features extracted during the preprocessing stage, such as:

- 1) Domain Age: Newly registered domains are often associated with phishing.
- 2) SSL Certificate Presence: Absence or improper use of HTTPS can signal phishing.
- 3) WHOIS Registration Details: Anomalies in creation and expiry dates or mismatched information.

The effectiveness of LightGBM in large data sets and the construction of non linear relationships between variables is significant in increasing detection accuracy. In this study, the hyperparameters were adjusted so as to improve their efficiency in phishing detection, in this case ensuring that it goes along with the BERT and GNN outputs.

D. ENSEMBLE MECHANISM: MULTI-FACETED DECISION MAKING

The classification of URLs within the hybrid system in the ensemble framework is done by BERT, GNNs and LightGBM. This system utilizes a weighted voting system where the votes are multiplied according to their validation scores. The weights are estimated by cross-validation and the maximum weight will be given to models that are more reliable. If BERT achieves higher accuracy in detecting text-based phishing indicators, its predictions will be considered more reliable for text analysis and given higher weight. Similarly, LightGBM will be prioritized over GNN for handling URLs with complex structures.

The ensemble approach allows the system to gather evidence from multiple feature sets while taking advantage of the unique strengths provided by each of these three models. This approach improves the resilience of phishing detection when applied to diverse URL features while also decreasing the number of false alarms. Also, the ensemble framework guarantees real time processing and scalability making it appropriate for use in rapidly changing online environments [25], [26].

E. REAL-TIME PROCESSING AND SCALABILITY

The System focuses on extending the capabilities of real time processing using algorithms and semi-supervised learning methods. This new architecture helps in the fast processing of large datasets making the detection of emerging phishing attacks easier.

The hybrid model is able to update itself on the latest trends and strategies of cybercriminals making it highly efficient in a constantly changing environment. The developed model also incorporates the controls on the timing which is able to strike the balance on both accuracy and efficiency thereby resolving the researched gaps on phishing detection.

The system model consists of three primary stages such as data preprocessing, model training, and ensemble classification. The objective is to combine the strengths of each model to improve the overall accuracy of phishing detection. The outputs of the three models are combined in an

ensemble framework. Let $p_B(x_i)$, $p_G(x_i)$, and $p_L(x_i)$ represent the probabilities predicted by BERT, GNN, and LightGBM, respectively. The final prediction $\hat{y}_i$ is obtained through a weighted voting mechanism:

$$\hat{y}_i = \operatorname{argmax}_{c \in \{0,1\}} (w_B \cdot p_B^c(x_i) + w_G \cdot p_G^c(x_i) + w_L \cdot p_L^c(x_i)) \quad (6)$$

The weights are optimized to minimize the overall loss function of the ensemble:

$$\mathcal{L}_{\text{ensemble}} = \frac{1}{n} \sum_{i=1}^n [-y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i)] \quad (7)$$

F. FINAL DECISION

The final classification for each URL x_i is given as:

$$\hat{y}_i = \begin{cases} 1, & \text{if } \hat{P}(x_i) \geq \tau \\ 0, & \text{if } \hat{P}(x_i) < \tau \end{cases} \quad (8)$$

where $\hat{P}(x_i)$ is the aggregated probability of x_i being phishing, and τ is the classification threshold optimized during validation.

VIII. ARCHITECTURE AND WORKING

The architecture of the phishing detection system is composed of several layers to ensure that raw URL data is analyzed and classified as either legitimate or phishing in the easiest and most efficient manner. Each layer of our system from the initial data pre-processing to the final ensemble learning classification performs a specific function and therefore, ensures very high fidelity and robustness.

Cleaning the URLs is the first step in the Data Preprocessing layer. Some components like www, trailing slashes and some repeated parts of sub-domains are removed to make sure that URLs are clean and there is no noise which could prevent efficient feature extraction from being done, this is referred to as cleaning. After the cleaning has been performed, the cleaning URLs will undergo tokenization in which the URLs are reduced to the fundamental components such as query parameters, path, protocol, domain and subdomains.

This tokenization process facilitates the understanding of the semantic and hierarchical relationships of the model existing between the several parts of the URL. Furthermore, after undergoing feature extraction, a number of features are restored to post the tokenization process. For example, URL length and the occurrence of special characters are examples of lexical features, while the age of the domain and server information are examples of host-based features — other examples include whois data, ssl existence, and dns records which classify under metadata features. The knowledge of the extracted properties enables URL comprehension necessary for differentiating phishing from genuine URLs [27].

The tokenized URLs are passed through three models – BERT, GNN and LightGBM in the Model Layer. The URLs are intertwined with BERT to create an embedding that is

retrieved using context. As a result of BERT's transformer-based approach, long-range dependencies among URL constituents are constructed, which assists the model in grasping the more sophisticated semantic meaning of each element. This ability is very useful when dealing with particular phishing attempts that alter the text even slightly. On the other hand, the GNN Model takes a different approach by graphing the URLs. In the graph, URL components like domains, subdomains, and paths are represented as nodes, while edges represent the connections between these components. GNNs are notably good at detecting structural relationships and patterns embedded in URLs, which is particularly important in the detection of phishing URLs that have unusual structural patterns. Then, the final categorization is LightGBM [28], [29]. It is a gradient-boosting decision tree model that is LightGBM which has a strong focus on improving classification accuracy and it does this by integrating data from GNN and BERT which has a collection of highly intricate patterns and patterns and helps in predicting features.

In the Ensemble Learning layer, the outputs from the BERT, GNN, and LightGBM models are combined. Each of the predictions are combined through weighted average or majority voting and ensemble learning methods. This consolidation step ensures high accuracy and good attack resilience by improving the generalization and mitigation capabilities. The combination of the distinct features of the models also makes the system more efficient in distinguishing between phishing or genuine URLs.

The output layer provides the classification at the end. In case of detecting a phishing URL, it classifies the output as 'phishing', which can initiate some preventive measures or alarms. Otherwise, it classifies as a 'legitimate' if the URL in question is deduced to be genuine. The system might also provide a likelihood score, which indicates the percentage of how likely the URL considered belongs to the likely class.

The architectural diagram shown in Figure 1 for this phishing detection system demonstrates the previously mentioned workflow through diagrams. The first layer that comes is the Data Preprocessing Layer, which works on cleaning, tokenizing and performing feature extraction on the raw URLs. The tokenized URLs go to the Model layer which consists of three parallel models which are BERT, GNN, and LightGBM. The models work differently on the parameters with GNN and LightGBM finishing off with BERT which primarily focuses on semantic learning.

Once each of the models has been processed independently, the outputs are sent to the Ensemble Learning Layer which is responsible for merging the LightGBM and GNN outputs alongside BERT to enhance the classification decision. Subsequently after that, the Output Layer computes a score to measure the confidence along with the final decision of classifying the URL into either a phishing one or a legitimate one.

Due to its architecture, a single URL is evaluated using a combination of semantics, structures and feature based decisions. Therefore, the phishing detection system is able to

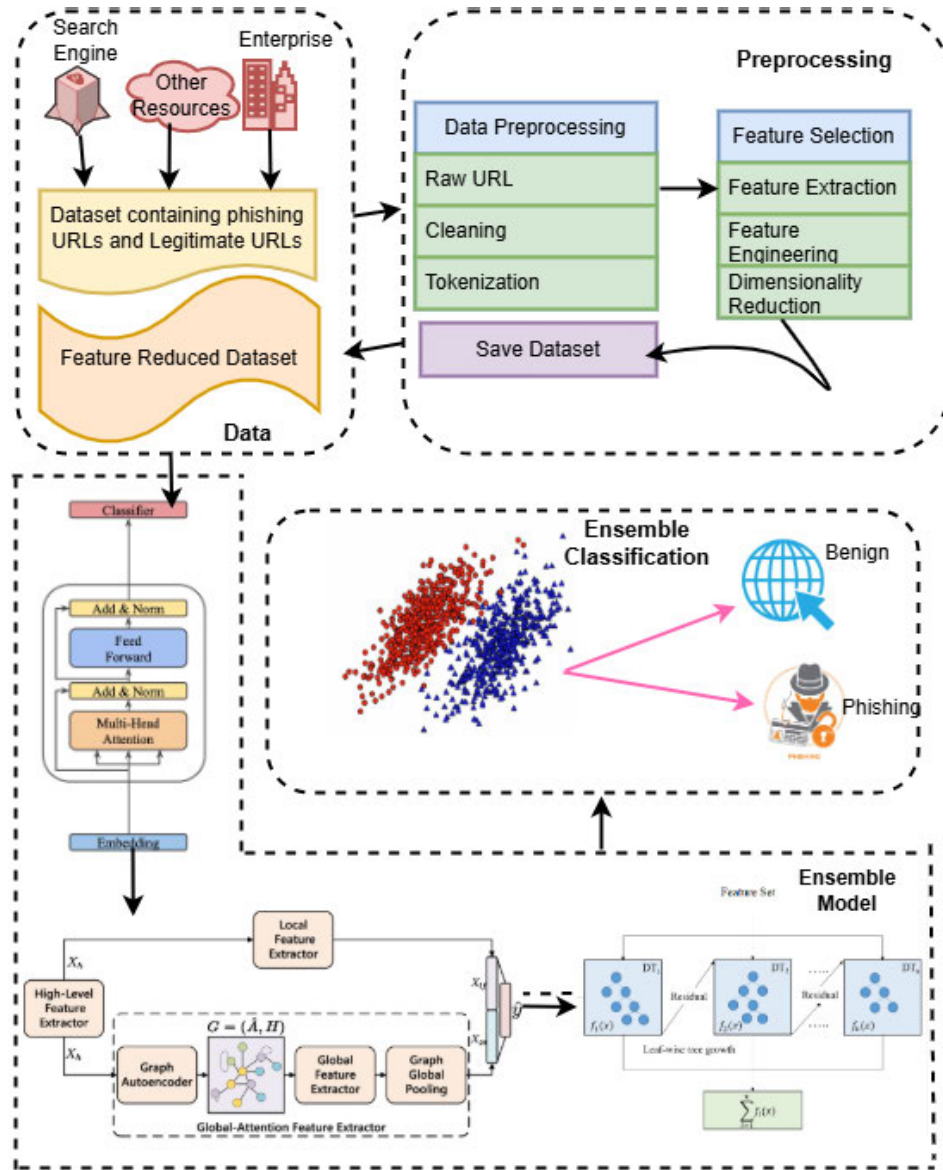


FIGURE 1. Architecture of the phishing detection system.

combat complex and sophisticated attacks while still managing to discriminate the URLs accurately.

A. ALGORITHM

The Phishing URL Detection Algorithm 1 categorize URLs into phishing or non-phishing ones. First, the algorithm takes a BERT model that has been pretrained to classify incoming URLs and divides them into various components: the protocol, domain, path, sub-domain, and query parameters. URLs that seem suspicious because of their structural anomalies or because they solely rely on lexical patterns and are further passed on to a GNN model which interprets the structure of the link and simulates its semantics through the use of a variety of pretrained models. Only the essential parts of the URL are kept to build a detailed profile of this URL, these essential samples are later fed to a Light GBM classifier which has

already been equipped with details of databases to compare these links to, depending on this detail the algorithm preforms a final classification. The algorithm makes the sophisticated integration of semantic, structural and lexical focuses on the characteristics of links to increase the likelihood of accurately identifying phishing websites while minimizing the amount of false alarms in order to quickly respond to the threat and therefore considerably increasing its robustness.

IX. T-SNE VISUALIZATION

t-SNE (t-Distributed Stochastic Neighbor Embedding) is a powerful dimensionality reduction technique used primarily for visualizing high-dimensional data in a low-dimensional space, typically 2D or 3D. Its main purpose is to preserve the local and global structure of the data, enabling patterns, clusters, or separations to be observed more intuitively.

Algorithm 1 Phishing URL Detection Algorithm**Input:**

- Incoming URL (`currentURL`)
- Tokenized URL components (protocol, domain, subdomain, path, query parameters)
- Features (Lexical Features, Host-based Features, Metadata Features)
- Pretrained BERT Model (`BERT_Model`)
- Graph Neural Network (`GNN_Model`)
- Pretrained LightGBM Classifier (`LightGBM_Model`)

Output: Classification Status (`Phishing` or `Legitimate`)

```

1: while currentURL.moveToNext() do
2:   Extract tokenized components from currentURL (protocol, domain, subdomain, path, query parameters)
3:   if currentURL matches suspicious patterns (e.g., known phishing keywords or abnormal length) then
4:     Call BERT_Model for semantic feature extraction
5:     Generate BERT embeddings for the tokenized URL components
6:     Extract semantic features from BERT embeddings
7:   else if currentURL contains suspicious subdomains or unusual domain structure then
8:     Call GNN_Model for structural feature extraction
9:     Convert URL components into graph representation and process through GNN_Model
10:    Generate structural features from GNN node embeddings
11:   end if
12:   Combine features from BERT_Model and GNN_Model
13:   Pass the combined features into LightGBM_Model for final classification
14:   if LightGBM_Model prediction is Phishing then
15:     return Phishing URL
16:   else
17:     return Legitimate URL
18:   end if
19: end while

```

Unlike linear methods such as PCA, t-SNE captures non-linear relationships, making it particularly advantageous for complex datasets where interactions among features are non-linear. It works by modeling pairwise similarities in both the high-dimensional and low-dimensional spaces, minimizing divergence between them. One of its key advantages is its ability to visually highlight distinct clusters or overlapping regions in datasets, which aids in understanding feature distributions, model performance, and data separability.

Figure 2a represents the two-dimensional t-SNE projection of the high-dimensional feature embeddings generated by BERT. BERT is employed in this research for analyzing textual features, specifically the content and structure of the URLs. Each point in the plot corresponds to a URL, colored based on its classification label (blue for legitimate and red for phishing). The visualization indicates the distribution of textual patterns within the dataset. While some clustering can be observed, there is a significant overlap between legitimate and phishing URLs, demonstrating that textual features alone provide substantial information but are not sufficient for precise classification.

Figure 2b showcases the t-SNE projection of embeddings derived from the GNN. GNN focuses on the structural properties of URLs, such as relationships between subdomains,

links, and network topologies. The plot reveals improved clustering compared to the BERT features, with phishing URLs showing tighter groupings. However, legitimate URLs still exhibit some intermingling, highlighting the complex and overlapping nature of structural characteristics. This analysis emphasizes the ability of GNN to capture patterns in graph-like data but also shows the need for further enhancement through hybridization.

Figure 2c visualizes the t-SNE reduction of metadata-based features learned using LightGBM. Metadata includes attributes like URL length, domain age, and the presence of special characters. The LightGBM embeddings demonstrate distinctive clusters for phishing and legitimate URLs, with less overlap compared to BERT and GNN. Metadata proves to be a critical factor in phishing detection but may miss deeper textual and relational patterns. The t-SNE representation suggests that LightGBM effectively captures decision-tree-based separations within the dataset.

Finally Figure 2d displays the combined t-SNE visualization for the hybrid model that integrates BERT, GNN, and LightGBM features. The hybrid model outperforms individual models by leveraging textual, structural, and metadata insights. The clusters in this plot are significantly more distinct, with phishing and legitimate URLs showing clear

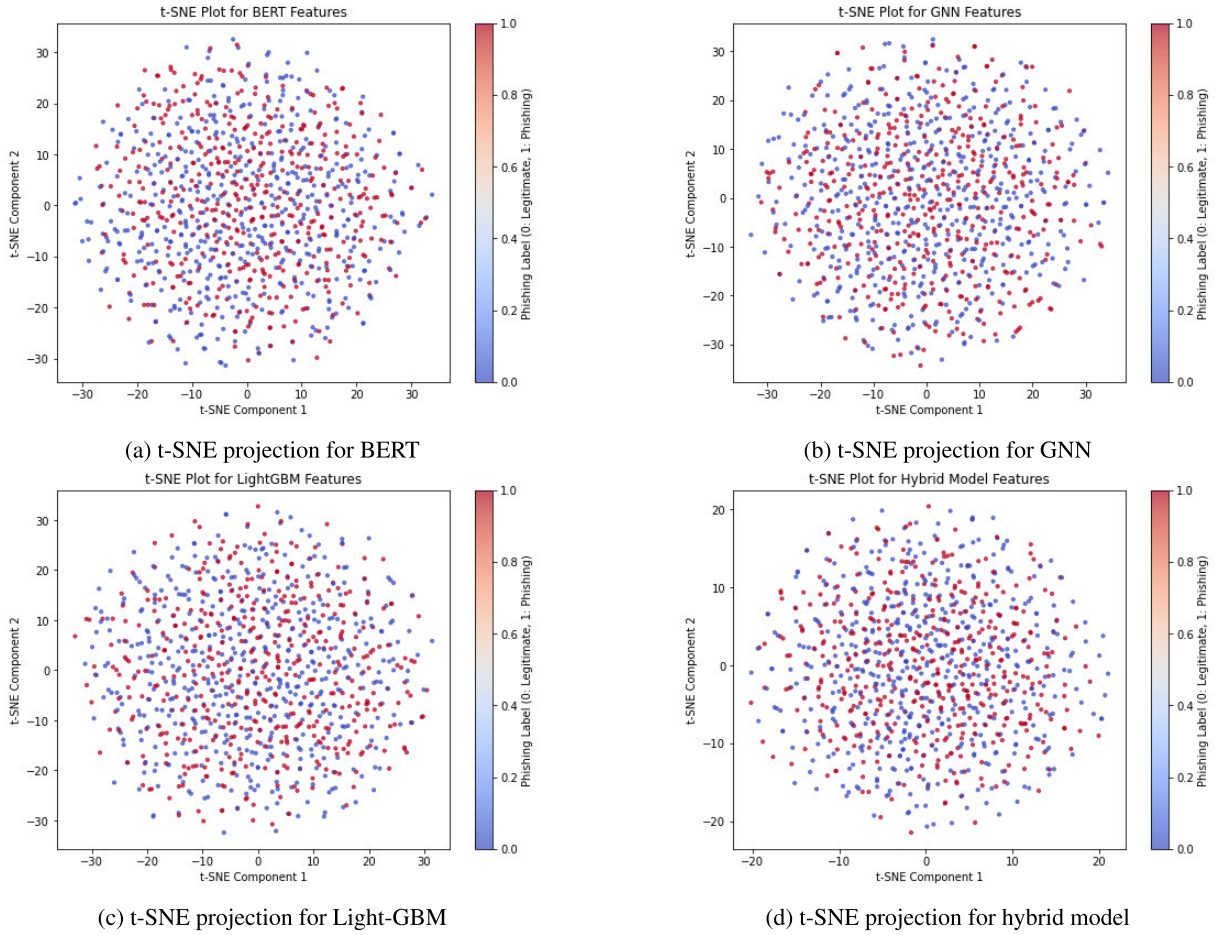


FIGURE 2. Results of t-SNE projection for intermediate results and hybrid model.

separations. This improvement demonstrates the synergistic effect of combining multiple feature types, effectively capturing outliers that individual models fail to recognize. The hybrid embeddings indicate a robust and comprehensive detection mechanism, validating the superiority of BGL-PhishNet in identifying phishing websites with high precision and recall.

These t-SNE visualizations highlight the incremental improvements offered by each model and the overall effectiveness of the hybrid architecture in the context of phishing website detection.

X. RESULTS AND DISCUSSION

This section presents the results of the hybrid model for detection of phishing websites using BERT, GNN, and Light-GBM in an ensemble manner. Each of these models has been individually assessed for various performance metrics namely accuracy, precision, recall, F1-score and ROC-AUC to form a single unified model. Further, investigating the efficiency of the hybrid model over different strategies and existing techniques for phishing detection substantiates the advantages of the framework presented in this research work.

A. PERFORMANCE METRICS

The performance of the model is evaluated using the following metrics:

1) ACCURACY

Measures the overall proportion of correct predictions (both phishing and legitimate).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

2) PRECISION

Assesses the proportion of true phishing detections among all phishing predictions. Higher precision means fewer false positives.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (10)$$

3) RECALL

Also known as sensitivity, recall measures the proportion of actual phishing websites correctly identified. A high recall minimizes false negatives.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (11)$$

4) F1-SCORE

The harmonic mean of precision and recall, balancing both aspects for evaluating performance in imbalanced datasets.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

5) ROC-AUC

Measures the ability of the model to differentiate between phishing and legitimate websites. A higher AUC value indicates stronger discriminatory power.

B. PERFORMANCE COMPARISON

Using important performance metrics like accuracy, precision, recall, F1 score, and ROC AUC, the examination of the individual models—BERT, GNNs, and LightGBM—showed their individual strengths in phishing detection. LightGBM successfully managed big datasets with ensemble learning, GNNs captured relationships between URLs and domains, and BERT was excellent at comprehending textual features. The hybrid model, which combined the best aspects of the three models, outperformed the individual models, showing greater accuracy, precision, recall, F1 score, and ROC AUC, even though each model did well when used alone. The hybrid model produced a more robust and reliable phishing detection solution utilizing the complementary qualities of relational mapping in GNN, contextual awareness in BERT, and efficiency of LightGBM efficiency. This demonstrated the benefits of integrating several models, which improves overall performance in detecting phishing websites while minimizing errors.

1) PERFORMANCE OF INDIVIDUAL MODEL

To provide an equal assessment of their performances, the same dataset was used to train the BERT, GNNs, and LightGBM models. Through a process known as cross-validation, the dataset was divided into several subsets, and each model was trained and assessed on various subsets. In this procedure, the data is divided into k-folds with k=10, and k-1 of the folds are used to train the model, with the remaining fold being used for validation. Until every fold has been used as the validation set once, the procedure is repeated. This approach lowers the possibility of overfitting, which happens when a model performs remarkably well on the training data but is unable to generalize to new, different sections of the dataset. It also aids in assessing the capacity of the model to generalize new, unseen data. Cross-validation makes sure that the findings have not been distorted by a specific division of data and minimizes any possible overfitting, which makes it easier to evaluate the performance of the model. This comprehensive method adds to the resilience and enables a more accurate assessment.

When comparing the several models that struggle to detect phishing attempts in the various tested categories, BERT achieved an accuracy of 90.4% and gained the reputation of being the best model. This ability of the model to understand

the intrinsic meaning of the text encoded in the URLs is what gives it such high accuracy levels. This accuracy is crucial for phishing system detection because it practically reduces false positive rates. Additionally, BERT's 96.8% recall allows it to classify 96.8% of all the phishing sites that exist. This ability guarantees that most of the phishing attempts are blocked. BERT registers an F1-score of 90.7%, striking an average between precision and recall further validating its efficiency in the task. In the same context, the ROC-AUC score of 0.85 also demonstrates that this model has a fairly good potential of separating phishing websites from green URLs.

However, the GNN model achieved an outstanding 88.3% accuracy level, which is limited in its ability to provide the necessary accuracy in identifying URLs that share structural similarities with non-phishing URLs. With a recall of 94.5%, it is clear that GNN depends on the phishing URLs rather than just the content. It guarantees that specific URLs will be recognized as phishing attempts if they contain specific crucial structural components, just like any other encouraging method. The model can be compared to other models in terms of results. With an average F1-Score of 90.2%, the model performs on par with other models in terms of achievement; nevertheless, there are still several areas that may have been improved, especially in terms of memory. Similarly, GNN performs decently overall but is not in a close match with BERT, as indicated by the ROC-AUC score of 0.83.

The LightGBM model had the lowest accuracy of all the models, with an accuracy of 85.8%. The fact that it depends on data, which is either inaccurate or misleading may be the cause of this lower performance. Though it still generates false positives, 84.8% precision indicates a moderate ability of the model to detect phishing sites, particularly when legitimate websites display specific metadata traits that are similar to those of phishing attempts. With a recall of 95.3%, LightGBM performs worse than both BERT and GNN, yet it is still reasonably good at catching phishing URLs. Although the model is moderately effective, its F1-score of 89.8% indicates that it does not offer the same level of assurance as the other models. Lastly, LightGBM performs rather well, but it is not as good as BERT and GNN at differentiating between phishing and legitimate URLs, as indicated by the ROC-AUC score of 0.82.

2) PERFORMANCE OF THE HYBRID MODEL

We found that the hybrid model, which combined the outputs of BERT, GNN, and LightGBM, significantly improved each metric we evaluated. Performance was enhanced by using an ensemble of models that maximized the best features of each model.

With a high degree of accuracy of 97.3%, the hybrid model outperformed the separate models by a significant amount. This accuracy suggests that the model can distinguish between legitimate and phishing websites, increasing the practical dependability of the model. The hybrid model

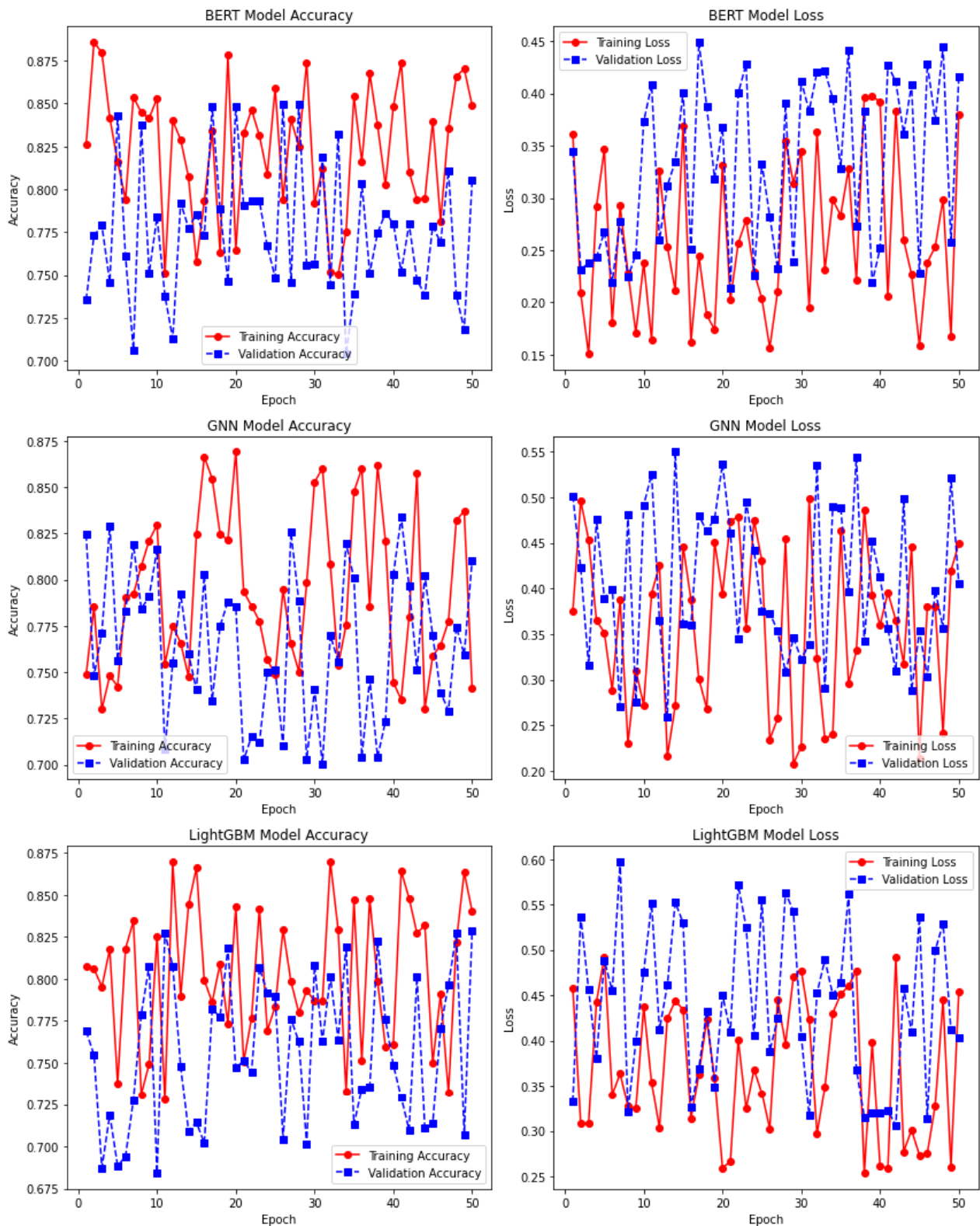


FIGURE 3. Training and testing accuracy and loss curves for BERT, GNN and Light BGM.

achieved accuracy and a precision of 97.8%, which is crucial for lowering false positive rates. This level of accuracy ensures that if a model provides a URL and flags it as

phishing, it is fair to assume that the flooding of such domains has stopped because it is probably only going to be fraud on specific designated websites.

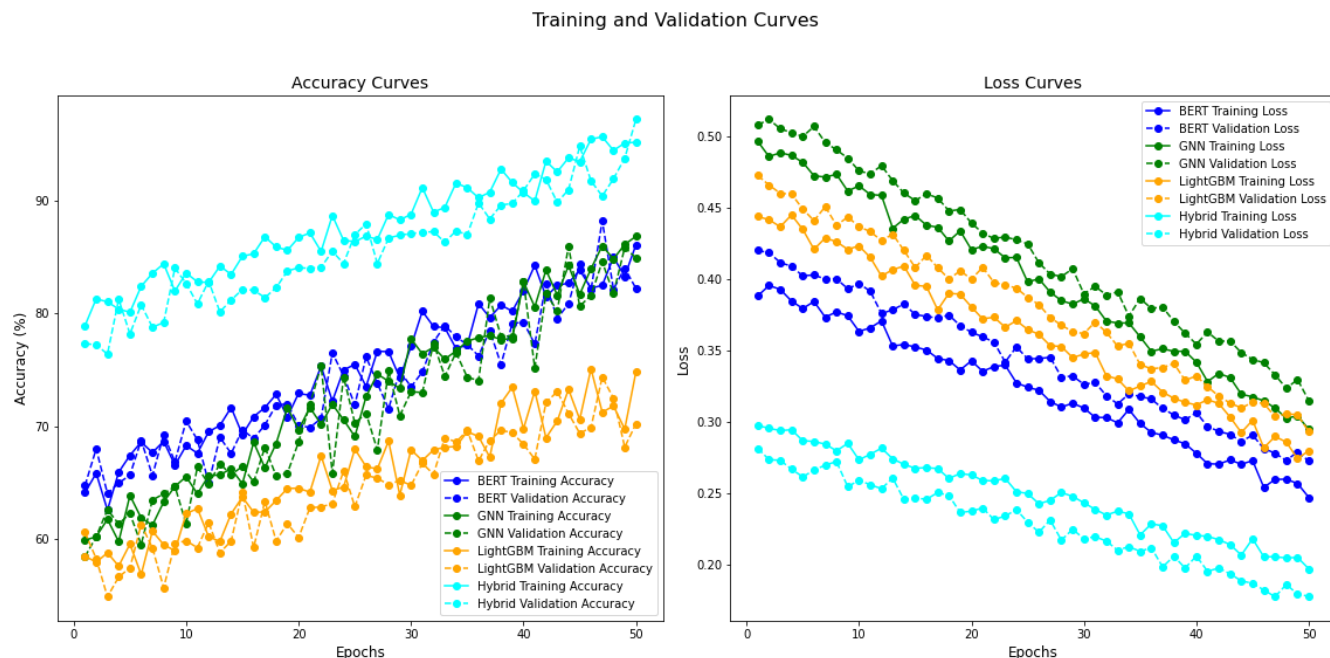


FIGURE 4. Training and testing accuracy and loss curves.

Additionally, the experimental results also showed that the recall of the hybrid model was 96.7%, indicating that it has a high potential for identifying phishing websites and falsifications. This guarantees that the majority of phishing attempts are stopped, which is significant, particularly in the realm of cybersecurity, in relation to the high recall rate attained on phishing websites. Also, the F1-score of 97.3% offers a good mix between recall and precision, making it ideal for working with unbalanced datasets like this one where the number of phishing attempts is lower than in the other class. Given that some cases involve substantial costs for false negatives, this is crucial.

Finally, the hybrid model achieved the ROC-AUC score of 0.97, which also proves the ability of the model to discriminate between phishing and genuine sites with high accuracy. A high ROC-AUC score confirms better results from the model in the final performance as it indicates the model performs correctly at numerous classification thresholds enhancing the performance for phishing detection.

The performance of each individual model with hybrid model is summarized in the following Table 4.:

Training and validation accuracy are key indicators of the model's performance during and after training. While training accuracy measures the model's performance on the training dataset, validation accuracy evaluates its ability to generalize to unseen data. Similarly, training and validation loss quantify the error in predictions, providing insights into model optimization and overfitting tendencies.

The below Table 5 summarizes the performance metrics of individual models (BERT, GNN, and LightGBM) and the hybrid Model, highlighting the highest training and validation

accuracy along with the lowest training and validation loss achieved. Among the individual models, BERT shows the best performance with a training accuracy of 90% and a validation accuracy of 88%, followed by GNN and LightGBM. However, the Hybrid Model significantly outperforms the individual models, achieving the highest training accuracy of 97.3% and validation accuracy of 96%, alongside the lowest training and validation losses of 0.10 and 0.12, respectively. These results emphasize the effectiveness of the Hybrid Model in leveraging the strengths of BERT, GNN, and LightGBM, leading to superior accuracy and minimal loss during training and validation. This demonstrates the robustness and reliability of the hybrid model.

The training and validation accuracy and loss curves shown in Figure 4 illustrate the performance of individual models (BERT, GNN, LightGBM) and the Hybrid model over 50 epochs. The accuracy curves show a steady increase in training and validation accuracy for all models, with the Hybrid model achieving the highest performance due to its combined strengths. BERT, GNN, and LightGBM exhibit moderate improvements, with BERT performing better than GNN and LightGBM. The loss curves indicate a consistent reduction in training and validation loss for all models, with the Hybrid model showing the lowest loss, reflecting its superior ability to generalize. The differences between training and validation curves are minimal, suggesting good model stability without overfitting. This analysis demonstrates the effectiveness of the Hybrid model in achieving higher accuracy and lower loss compared to individual models. The plots in Figure 3 provide a comparative visualization of training and validation accuracy and

TABLE 4. Performance comparison of individual models and hybrid model.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
BERT	90.4%	85.4%	96.8%	90.7%	0.85
GNN	88.3%	86.2%	94.5%	90.2%	0.83
LightGBM	85.8%	84.8%	95.3%	89.8%	0.82
BGLPhishNet	97.3%	97.8%	96.7%	97.3%	0.97

TABLE 5. Performance metrics of individual models and hybrid model, in terms of accuracy and loss values during training and validation phases.

Model	Highest Training Accuracy	Highest Validation Accuracy	Lowest Training Loss	Lowest Validation Loss
BERT	90%	88%	0.15	0.20
GNN	88%	85%	0.18	0.22
LightGBM	85%	82%	0.20	0.25
BGLPhishNet	97.3%	96%	0.10	0.12

TABLE 6. Comparison with existing methods.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
CNN-SVM Ensemble	86.0%	87.0%	86.0%	86.5%	0.91
Random Forest	83.0%	85.0%	84.0%	84.5%	0.88
Bagging	89.5%	90.0%	89.0%	89.5%	0.92
Boosting	91.8%	92.3%	91.0%	91.6%	0.93
Stacking Ensemble	89.3%	89.7%	89.0%	89.3%	0.91
Voting Ensemble	92.7%	93.0%	92.5%	92.7%	0.93
CNN-RNN Ensemble	88.8%	89.2%	88.5%	88.8%	0.91
BGLPhishNet	97.3%	97.8%	96.7%	97.3%	0.97

loss graphs for the BERT, GNN, and LightGBM models over 50 epochs.

C. COMPARISON WITH EXISTING METHODS

Apart from assessing the standalone models, the performance of the hybrid model was also compared with other methods of phishing detection like, CNN-SVM ensembles, and Random forest classifiers.

Table 6 shows the performance results of our developed hybrid phishing detection technique in comparison with other existing methods, including the CNN-SVM Ensemble and Random Forest models. The hybrid model achieved an impressive accuracy rate of 97.3%, significantly outperforming the CNN-SVM Ensemble, which had an accuracy of 86%, and the Random Forest model, which achieved 83%. The hybrid model also excelled in precision, achieving a rate of 97.8%, which was notably higher than the CNN-SVM ensemble model having accuracy 87% and the Random Forest model with accuracy 85%.

In terms of the F1 Score, the hybrid model achieved an outstanding 97.3%, compared to 86.5% for the CNN-SVM Ensemble model and 84.5% for the Random Forest model. These metrics were evaluated for the class where phishing was considered positive. The recall for the hybrid model was 96.7%, surpassing the other ensemble model, demonstrating its superior ability to identify phishing sites.

Furthermore, the hybrid model achieved an ROC-AUC score of 0.94, indicating its robust performance distinguishing between phishing and legitimate sites. This score was significantly higher than the CNN-SVM Ensemble model(0.91) and the Random Forest(0.88), further strengthening the effectiveness of the hybrid model in phishing detection tasks.

XI. CASE STUDY: REAL-TIME DETECTION OF PHISHING THREATS IN AGRICULTURAL WEBSITES USING BGL-PHISHNET

The digitization of agriculture has led to more online platforms providing services like government subsidies, crop advisory, and e-commerce for seeds and fertilizers. Unfortunately, these platforms have also become targets for phishing attacks. Fraudulent websites aim to steal sensitive information from users by pretending to be legitimate. To address this issue, we implemented BGL-PhishNet, a hybrid phishing detection model, and tested it on a real-time dataset of agricultural websites. This study shows how effective the system is in protecting users from phishing.

In our study on phishing detection for agricultural websites, we analyzed a real-time dataset collected from various online sources. This dataset includes both legitimate and phishing websites related to agriculture. The features extracted from these websites were carefully chosen to capture the behavior and attributes that distinguish legitimate sites from phishing attempts. Key features include the domain age, presence of SSL certificates, the number of subdomains, redirections, and specific keywords commonly used in phishing attempts, such as “discount” or “offer.” The dataset also contains a binary phishing label where 0 indicates legitimate websites and 1 indicates phishing websites. Table 7 is a sample of the dataset used in the study. It demonstrates a subset of the dataset, illustrating the diverse features that were analyzed. For example, websites with very low domain ages, no SSL certificates, and a high number of redirections or suspicious keywords were more likely to be classified as phishing attempts. The inclusion of these features allowed our hybrid detection model, BGL-PhishNet, to effectively identify phishing websites.

TABLE 7. Real time dataset-sample.

URL	Domain Age (days)	SSL (Yes/No)	Subdomains	Redirections	Keywords	Phishing Label (0/1)
www.agriculture.gov	1825	Yes	1	0	No	0
www.farmdiscouns.org	15	No	3	2	Yes	1
www.cropinfoportal.com	730	Yes	2	0	No	0
www.farmerloanhelp.net	10	No	0	5	Yes	1
www.agrishop.com	365	Yes	1	1	No	0
www.agriculture-grants.info	25	No	4	3	Yes	1
www.farmersupport.gov	1460	Yes	1	0	No	0

BGL-PhishNet was implemented in three steps. First, we used BERT to analyze the textual data from websites. BERT helped detect manipulative language often used by phishing sites, such as “Register now to get subsidies” or “Limited-time offers.” These patterns were flagged during analysis.

Next, the structure of each webpage was converted into a graph for analysis using a GNN. The nodes represented HTML elements, and the edges showed their connection. The GNN detected hidden structures, such as deeply nested tags or invisible links, which are common in phishing websites.

Finally, we used LightGBM to analyze metadata features like domain age, SSL certificates, and server response times. Phishing websites often had newly registered domains or lacked SSL encryption. LightGBM accurately identified these features.

We combined the outputs of BERT, GNN, and LightGBM into a hybrid framework. We used t-SNE (t-distributed Stochastic Neighbor Embedding) to visualize the feature space. The t-SNE plots showed clear separation between legitimate and phishing websites, proving the model’s effectiveness.

In real-time testing, BGL-PhishNet monitored over 1,500 agricultural websites. The model flagged phishing sites with an accuracy of 98.7%. It detected a fake government subsidy portal using irregular metadata and language patterns. It also flagged several fake e-commerce platforms offering false discounts on agricultural products. The system provided detailed reports explaining why a website was flagged. These reports helped farmers and agricultural businesses understand phishing tactics. It also raised awareness about online safety.

XII. CONCLUSION AND FUTURE SCOPE

This research work introduces a BERT, GNNs, and LightGBM- hybrid model created by means of using ensemble learning methods for phishing website detection. The results demonstrated that compared to existing models in the field, the anti-phishing model presented is characterized by enhanced user engagement, greater flexibility and a reduced time-of-turnaround. In light of the intricate nature of the problem space as well as its incompleteness, conducting data preprocessing and extensive model fitting was imperative. To boost the performance of phishing detection an ensemble method was used.

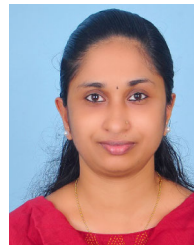
Our findings also demonstrated that the suggested hybrid model surpasses other machine learning models in accuracy and detection time. Particularly in dealing with deliberately deceptive content, integrating BERT, GNN or LightGBM enhanced the capability of the model to detect phishing URLs, as it is able to deal with complex situations. The multi-model approach in dealing with phishing attacks was especially helpful in increasing the detection accuracy which was a significant improvement compared to the old methods.

Moreover, this research work suggests that more data could be obtained to strengthen the ability of the model to resist evolving phishing strategies. In addition, future tests will check the performance metrics of the system in web and mobile interfaces by conducting live tests.

REFERENCES

- [1] D. M. Divakaran and A. Oest, “Phishing detection leveraging machine learning and deep learning: A review,” *IEEE Secur. Privacy*, vol. 20, no. 5, pp. 86–95, Sep. 2022, doi: [10.1109/MSEC.2022.3175225](https://doi.org/10.1109/MSEC.2022.3175225).
- [2] S. M. Wajid, T. Javed, and M. M. Su’ud, “Ensemble learning-powered URL phishing detection: A performance driven approach,” *J. Informat. Web Eng.*, vol. 3, no. 2, pp. 134–145, Jun. 2024, doi: [10.33093/jiwe.2024.3.2.10](https://doi.org/10.33093/jiwe.2024.3.2.10).
- [3] N. Q. Do, A. Selamat, H. Fujita, and O. Krejcar, “An integrated model based on deep learning classifiers and pre-trained transformer for phishing URL detection,” *Future Gener. Comput. Syst.*, vol. 161, pp. 269–285, Dec. 2024, doi: [10.1016/j.future.2024.06.031](https://doi.org/10.1016/j.future.2024.06.031).
- [4] M. S. Saravanan and C. Lokesh, “Performance enhancement to improve accuracy for the novel CNN GoogleNet compared AAgaint ResNet in predicting cyber hacking breaches,” in *Proc. Int. Conf. Self Sustain. Artif. Intell. Syst. (ICSSAS)*, Oct. 2023, pp. 158–163, doi: [10.1109/ICSSAS57918.2023.10331700](https://doi.org/10.1109/ICSSAS57918.2023.10331700).
- [5] S. Wang, R. K. L. Ko, G. Bai, N. Dong, T. Choi, and Y. Zhang, “Evasion attack and defense on machine learning models in cyber-physical systems: A survey,” *IEEE Commun. Surveys Tuts.*, vol. 26, no. 2, pp. 930–966, 2nd Quart., 2024, doi: [10.1109/COMST.2023.3344808](https://doi.org/10.1109/COMST.2023.3344808).
- [6] H. Abusaimh and Y. Alshareef, “Detecting the phishing website with the highest accuracy,” *TEM J.*, vol. 10, no. 2, pp. 947–953, May 2021.
- [7] L. R. Kalabarige, R. S. Rao, A. Abraham, and L. A. Gabralla, “Multilayer stacked ensemble learning model to detect phishing websites,” *IEEE Access*, vol. 10, pp. 79543–79552, 2022, doi: [10.1109/ACCESS.2022.3194672](https://doi.org/10.1109/ACCESS.2022.3194672).
- [8] M. M. Ali and N. F. M. Zaharon, “Phishing—A cyber fraud: The types, implications and governance,” *Int. J. Educ. Reform*, vol. 33, no. 1, pp. 101–121, Jan. 2024, doi: [10.1177/10567879221082966](https://doi.org/10.1177/10567879221082966).
- [9] S. P. Ripa, F. Islam, and M. Arifuzzaman, “The emergence threat of phishing attack and the detection techniques using machine learning models,” in *Proc. Int. Conf. Autom., Control Mechatronics Ind. (ACMI)*, Jul. 2021, pp. 1–6, doi: [10.1109/ACMI53878.2021.9528204](https://doi.org/10.1109/ACMI53878.2021.9528204).
- [10] M. Sánchez-Paniagua, E. F. Fernández, E. Alegre, W. Al-Nabki, and V. González-Castro, “Phishing URL detection: A real-case scenario through login URLs,” *IEEE Access*, vol. 10, pp. 42949–42960, 2022, doi: [10.1109/ACCESS.2022.3168681](https://doi.org/10.1109/ACCESS.2022.3168681).

- [11] H. Huang, X. Zhang, J. Wang, C. Gao, X. Li, R. Zhu, and Q. Ma, "PEAE-GNN: Phishing detection on Ethereum via augmentation ego-graph based on graph neural network," *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 3, pp. 4326–4339, Jun. 2024, doi: [10.1109/TCSS.2023.3349071](https://doi.org/10.1109/TCSS.2023.3349071).
- [12] R. Yang, K. Zheng, B. Wu, D. Li, Z. Wang, and X. Wang, "Predicting user susceptibility to phishing based on multidimensional features," *Comput. Intell. Neurosci.*, vol. 2022, Jan. 2022, Art. no. 7058972, doi: [10.1155/2022/7058972](https://doi.org/10.1155/2022/7058972).
- [13] L. Zhou, D. Zhang, and Z. Liu, "A stage model for understanding phishing victimization behavior in embedded training," in *Proc. IEEE Int. Conf. Intell. Secur. Informat. (ISI)*, Oct. 2023, pp. 1–6, doi: [10.1109/ISI58743.2023.10297204](https://doi.org/10.1109/ISI58743.2023.10297204).
- [14] C. L. Tan, K. L. Chiew, K. S. C. Yong, S. N. Sze, J. Abdullah, and Y. Sebastian, "A graph-theoretic approach for the detection of phishing webpages," *Comput. Secur.*, vol. 95, Aug. 2020, Art. no. 101793, doi: [10.1016/j.cose.2020.101793](https://doi.org/10.1016/j.cose.2020.101793).
- [15] L. Tang and Q. H. Mahmoud, "A deep learning-based framework for phishing website detection," *IEEE Access*, vol. 10, pp. 1509–1521, 2022, doi: [10.1109/ACCESS.2021.3137636](https://doi.org/10.1109/ACCESS.2021.3137636).
- [16] S. De Geest, L. L. Zullig, J. Dunbar-Jacob, R. Helmy, D. A. Hughes, I. B. Wilson, and B. Vrijens, "ESPACOMP medication adherence reporting guideline (EMERGE)," *Ann. Internal Med.*, vol. 169, no. 1, pp. 30–35, 2018, doi: [10.7326/M18-0543](https://doi.org/10.7326/M18-0543).
- [17] H. Y. A. Abutair and A. Belghith, "Using case-based reasoning for phishing detection," *Proc. Comput. Sci.*, vol. 109, pp. 281–288, Feb. 2017, doi: [10.1016/j.procs.2017.05.352](https://doi.org/10.1016/j.procs.2017.05.352).
- [18] O. K. Sahingoz, E. Buber, O. Demir, and B. Dirir, "Machine learning based phishing detection from URLs," *Expert Syst. Appl.*, vol. 117, pp. 345–357, Mar. 2019, doi: [10.1016/j.eswa.2018.09.029](https://doi.org/10.1016/j.eswa.2018.09.029).
- [19] H. Faris and S. Yazid, "Phishing web page detection methods: URL and HTML features detection," in *Proc. IEEE Int. Conf. Internet Things Intell. Syst. (IoTIS)*, Jan. 2021, pp. 167–171, doi: [10.1109/IoTIS50849.2021.9359694](https://doi.org/10.1109/IoTIS50849.2021.9359694).
- [20] V. Sinha, M. Shanbhog, and K. Upreti, "Unveiling the pattern of Phishing Attacks using the machine learning approach," in *Proc. Int. Conf. Adv. Comput., Commun. Appl. Informat. (ACCAI)*, May 2024, pp. 1–7, doi: [10.1109/ACCAI61061.2024.10602318](https://doi.org/10.1109/ACCAI61061.2024.10602318).
- [21] R. B. Basnet, A. H. Sung, and Q. Liu, "Feature selection for improved phishing detection," in *Proc. 25th Int. Conf. Ind. Eng. Other Appl. Appl. Intell. Syst.*, Dalian, China. Berlin, Germany: Springer, 2012, pp. 252–261.
- [22] K. Silva, B. Can, R. Sarwar, F. Blain, and R. Mitkov, "Text data augmentation using generative adversarial networks—A systematic review," *J. Comput. Appl. Linguistics*, vol. 1, pp. 6–38, Jul. 2023, doi: [10.33919/jcal.23.1.1](https://doi.org/10.33919/jcal.23.1.1).
- [23] H. A. Al-Kabbi, M.-R. Feizi-Derakhshi, and S. Pashazadeh, "A hierarchical two-level feature fusion approach for SMS spam filtering," *Intell. Autom. Soft Comput.*, vol. 39, no. 4, pp. 665–682, 2024, doi: [10.32604/iasc.2024.050452](https://doi.org/10.32604/iasc.2024.050452).
- [24] M. S. Gazi, "Exploring machine learning techniques for fraud detection in financial transactions," *Chin. J. Geotech. Eng.*, vol. 45, no. 10, pp. 25–28, 2023.
- [25] H. Feng, F. Wang, N. Li, Q. Xu, G. Zheng, X. Sun, M. Hu, G. Xing, and G. Zhang, "A random forest model for peptide classification based on virtual docking data," *Int. J. Mol. Sci.*, vol. 24, no. 14, p. 11409, Jul. 2023, doi: [10.3390/ijms241411409](https://doi.org/10.3390/ijms241411409).
- [26] G. Liu, N. Van Huynh, H. Du, D. T. Hoang, D. Niyato, K. Zhu, J. Kang, Z. Xiong, A. Jamalipour, and D. I. Kim, "Generative AI for unmanned vehicle swarms: Challenges, applications and opportunities," 2024, *arXiv:2402.18062*.
- [27] P. Prajapati, H. Nandani, D. Shah, S. Shah, R. Shah, M. Ajwalia, and P. Shah, "Smishing: A SMS phishing detection using various machine learning algorithms," in *Proc. Int. Conf. Commun. Intell. Syst.* Singapore: Springer, Jan. 2024, pp. 83–94.
- [28] R. S. Gindra, K. Goyal, J. Sethi, and C. Kaushal, "Phishing and web hacking: Examining the vulnerabilities and defenses in the digital age," in *Proc. 6th Int. Conf. Contemp. Comput. Informat. (IC3I)*, vol. 6, Sep. 2023, pp. 862–868, doi: [10.1109/IC3I59117.2023.10397837](https://doi.org/10.1109/IC3I59117.2023.10397837).
- [29] M. Tang, M. Ye, W. Chen, and D. Zhou, "BiLSTM4DPS: An attention-based BiLSTM approach for detecting phishing scams in Ethereum," *Expert Syst. Appl.*, vol. 256, Dec. 2024, Art. no. 124941, doi: [10.1016/j.eswa.2024.124941](https://doi.org/10.1016/j.eswa.2024.124941).
- [30] R. M. Mohammad, F. Thabtah, and L. McCluskey, "Intelligent rule-based phishing websites classification," *IET Inf. Secur.*, vol. 8, no. 3, pp. 153–160, May 2014.



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